

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

Weight (pounds)	Frequency
13	12
14	8
15	5
16	7
17	9
18	10
19	13
20	7

The frequency table summarizes a data set of the weights, rounded to the nearest pound, of **71** tortoises. A weight of **39** pounds is added to the original data set, creating a new data set of the weights, rounded to the nearest pound, of **72** tortoises. Which statement best compares the mean and median of the new data set to the mean and median of the original data set?

Answer

- A. The mean of the new data set is greater than the mean of the original data set, and the median of the new data set is greater than the median of the original data set.
- B. The mean of the new data set is greater than the mean of the original data set, and the medians of the two data sets are equal.
- C. The mean of the new data set is less than the mean of the original data set, and the median of the new data set is less than the median of the original data set.
- D. The mean of the new data set is less than the mean of the original data set, and the medians of the two data sets are equal.

Correct Answer: B

Rationale

Choice B is correct. The mean of a data set is the sum of the values in the data set divided by the number of values in the data set. The new data set consists of the weights of the **71** tortoises in the original data set and one additional weight, **39**. Since the additional weight, **39**, is greater than any of the values in the original data set, the mean of the new data set is greater than the mean of the original data set. If a data set contains an odd number of data values, the median is represented by the middle data value in the list when the data values are listed in ascending or descending order. Since the original data set consists of the weights of **71** tortoises and is in ascending order, the median of the original data set is represented by the middle value, or the **36th** value. Based on the frequencies shown in the table, the **36th** value in this data set is **17**. If a data set contains an even number of data values, the median is between the two middle data values when the values are listed in ascending or descending order. Since the new data set consists of the weights of **72** tortoises, the median of the new data set is between the **36th** and **37th** data values when the values are arranged in ascending order. To keep the data in ascending order, the additional value of **39** would be placed at the bottom of the frequency table with a frequency of **1**. Therefore, based on the frequencies in the table, the **36th** and **37th** values in the new data set are both **17**. It follows that the median of the new data set is **17**, which is the same as the median of the original data set. Therefore, the mean of the new data set is greater than the mean of the original data set, and the medians of the two data sets are equal.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.